

The **tautological 1-form** is a 1-form $\alpha \in \Lambda^1 T^*T^*X$, for a manifold X . Its differential $\omega = d\alpha$ is a symplectic form. We now define α :

Let

$$\begin{array}{ccc} T^*X & & T_{(x,z_x)}T^*X \\ \uparrow \nearrow \pi & \rightsquigarrow & \downarrow d\pi \\ X & & T_x X \xrightarrow{z_x} \mathbb{R}. \end{array}$$

We define α pointwise. At $(x, z_x) \in T^*X$,

$$\alpha_{(x, z_x)} : T_{(x, z_x)}T^*X \longrightarrow \mathbb{R}, \quad \alpha_{(x, z_x)} = z_x \circ d\pi = \pi^* z_x.$$

Note that if we extend z_x to a section $\zeta: X \longrightarrow T^*X$ then

$$(\zeta^* \alpha)_x = \alpha_{(x, z_x)}(d\zeta|_x) = z_x \circ d\pi \circ d\zeta_x = z_x \circ d(\underbrace{\pi \circ \zeta_x}_{id}) = z_x$$

Let $\omega = d\alpha$ be the associated symplectic form. We say that a submanifold of a symplectic manifold is **isotropic** if the restriction of the symplectic form vanishes.

Let $\zeta: X \longrightarrow T^*X$ be a section. When is $\text{Im } \zeta$ isotropic? ← diffeomorphic to X

$$\text{Im } \zeta \text{ is isotropic iff } \zeta^* d\alpha = 0,$$

$$\text{iff } d\zeta^* \alpha = 0,$$

$$\text{iff } d\zeta = 0. \quad (\text{by the above } \zeta^* \alpha = \zeta) \quad \leftarrow \text{inclusion as submanifold}$$

Note: If the isotropic submanifold has maximal dimension, i.e. half the dimension of the ambient space, then we call it **Lagrangian**. Hence, images of sections are Lagrangian iff their differentials vanish.

Why are Lagrangian submanifolds interesting? Let $(M_1, \omega_1), (M_2, \omega_2)$ be symplectic manifolds. Let

$$\begin{array}{ccc} M_1 \times M_2 & & \\ \pi_1 \swarrow & \searrow \pi_2 & \\ M_1 & & M_2. \end{array}$$

Then $M_1 \times M_2$ is symplectic with symplectic form $\omega = \pi_1^* \omega_1 - \pi_2^* \omega_2$. Let $\varphi: M_1 \longrightarrow M_2$ be a diffeomorphism. Let

$$\Gamma_\varphi = \{(m_1, m_2) \in M_1 \times M_2 : m_2 = \varphi(m_1)\} \subset M_1 \times M_2$$

be the graph of φ . Then

$$\Gamma_\varphi \subset M_1 \times M_2 \text{ is Lagrangian iff } (id \times \varphi)^* \omega = 0,$$

$$\text{iff } \underbrace{(id \times \varphi)^* \pi_1^* \omega_1}_{id^*} - \underbrace{(id \times \varphi)^* \pi_2^* \omega_2}_{\varphi^*} = 0,$$

$$\text{iff } \omega_1 - \varphi^* \omega_2 = 0,$$

i.e. Γ_φ is Lagrangian iff φ is a symplectomorphism.